

Lab 6 – 03.06.2026

During the class, we will follow a course that has been proposed by Google:

<https://developer.android.com/courses/android-basics-compose/course>

Tasks for the sixth lab:

1. Unit 6: Data Persistence

<https://developer.android.com/courses/android-basics-compose/unit-6>

2. Pathway 1: Introduction to SQL

<https://developer.android.com/courses/pathways/android-basics-compose-unit-6-pathway-1>

Learn how to use SQL to read and manipulate data in a relational database.

- a. Use SQL to read and write to a database

<https://developer.android.com/codelabs/basic-android-kotlin-compose-sql>

3. Pathway 2: Use Room for Data Persistence

<https://developer.android.com/courses/pathways/android-basics-compose-unit-6-pathway-2>

Use the Room library to easily create and use relational databases in an Android app.

- a. Persist Data with Room

<https://developer.android.com/codelabs/basic-android-kotlin-compose-persisting-data-room>

- b. Read and update data with Room

<https://developer.android.com/codelabs/basic-android-kotlin-compose-update-data-room>

- c. Practice: Build Bus Schedule app (**mandatory**)

<https://developer.android.com/codelabs/basic-android-kotlin-compose-practice-bus-schedule-app>

4. Pathway 3: Store and Access Data Using Keys with DataStore

<https://developer.android.com/courses/pathways/android-basics-compose-unit-6-pathway-3>

Learn how to store simple, key-value pair data with Preferences DataStore in an Android app.

- a. Save preferences locally with DataStore

<https://developer.android.com/codelabs/basic-android-kotlin-compose-datastore>

- b. Project: Create a flight search app (**mandatory**)
<https://developer.android.com/codelabs/basic-android-kotlin-compose-flight-search>

As a result of the work please take the Quiz at the end of each Pathway.

Submission requirements:

- Use the **private** repository named MAD created during Lab 2.
- Inside the repository maintain the following folder structure:

```
MAD/  
  UNIT6/  
    BusScheduleApp      (Practice: Build Bus Schedule app)  
    FlightSearchApp/    (Project: Create a flight search app)
```

- The instructor should already be a collaborator (rmaciaszczyk@zut.edu.pl). If not, add them using that email address.
- Submit the repository link as a response to this assignment in **Microsoft Teams**.